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P6-1

3. $C_6H_{12} \rightleftharpoons 3H_2 + C_6H_6$

$A \rightleftharpoons 3B + C$ $K_c = 0.001 \left(\frac{mol}{dm^3}\right)^3$

$$\frac{dF_A}{dV} = r_A$$

$$\frac{dF_B}{dV} = 3r_B - R_B \qquad R_B = k_c C_B$$

$$\frac{dF_C}{dV} = r_C$$

$$r_A = -k\left(C_A - \frac{C_B^3 C_C}{K_c}\right)$$

$$C_A = C_{T0}\frac{F_A}{F_T}$$

$$C_B = C_{T0}\frac{F_B}{F_T}$$

$$C_C = C_{T0}\frac{F_C}{F_T}$$

$$-r_A = kC_{T0}\left(\left(\frac{F_A}{F_T}\right) - \frac{C_{T0}}{K_c}\left(\frac{F_B}{F_T}\right)^3\left(\frac{F_C}{F_T}\right)\right)$$

$$F_T = F_A + F_B + F_C$$

- more hydrogen is produced

4.

1) varying C_{B0}, v_0, V_0

- C_{B0} - rate increases, conversion increases, concentration of C_B increases, concentration of C_A decreases, concentration of C_D increases
- v_0 - rate increase, conversion increases, concentration of C_A decreases, concentration of C_B increases, concentration of C_D decreases.

- V_0 - conversion decreases, rate decreases, concentration of C_A increases, concentration of C_B decreases, concentration of C_D decreases

2) - increasing V_0 decreases concentration so reaction rate decreases
- higher v_0 increases C_A which increases reaction rate
- increasing C_{B0} increases rate because

$$r_A = -kC_AC_B$$

$$C_B = C_{T0}\left(\frac{F_B}{F_T}\right)$$

higher C_{B0} means higher C_B so r_A increases
C_A slows down r_A when too much C_B

3) $A + B \rightleftharpoons C + D$

$$-r_A = k\left(C_AC_B - \frac{C_CC_D}{K_c}\right)$$

- increases reaction rate, conversion decreases

5.
1). α has the greatest effect on conversion

2) $p = 0.1$ Catalyst weight = 80 kg $C_{A0} = 0.046$
$C_{B0} = 0.046$

$p = \frac{P}{P_0} \Rightarrow 0.1 = \frac{P}{P_0}$ $P = 1\,atm$

$$\frac{dp}{dW} = \frac{-\alpha}{2p}\left(\frac{F_T}{F_{T0}}\right)\left(\frac{T}{T_0}\right)$$

$$C_A = \frac{F_A}{v_0}$$

$$C_A = C_{T0}\left(\frac{F_A}{F_T}\right)p$$

$p = 0.1$ at high α - not included in graph

- higher α
- larger catalyst weight

3) - α reduces pressure so large drop in pressure

- higher catalyst weight decreases pressure

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semibatch reactor

$$CH_3COOC_2H_5 + C_4H_9OH \rightleftharpoons CH_3COOC_4H_9 + C_2H_5OH$$

$$A + B \rightleftharpoons C + D$$

$T = 300K$

$K_c = 1.08$

$-r_A = 9 \cdot 10^{-5} \frac{dm^3}{mol \cdot s}$

$V_0 = 200\ dm^3$

$v_0 = 0.05 \frac{dm^3}{s}$

$C_{B,F} = 10.93 \frac{mol}{dm^3}$

$C_{A0} = 7.72 \frac{mol}{dm^3}$

3. $C_D \approx 0$

$$-r_A = k\,(C_A C_B)$$

$$-r_A = k\left(C_A C_B - \frac{C_C C_D}{K}\right)$$

4. varying

F_{B0} – conversion of A increases

N_{A0} – decreases conversion

v_0 – conversion decreases, rate decreases

6. How would results change if reaction was reversible?

- requires understanding reversible vs irreversible rates.